

**INTERSECONDARY SCHOOLS EXAMINATION SERIES
ISESE**



**FORM TWO PRE-NECTA SERIES No.01
CHEMISTRY**

032

TIME 2:30 HRS

Year 2026

INSTRUCTIONS

1. This paper consists of sections A and B with a total of **ten (10)** questions.
2. Answer **all** questions in the spaces provided.
3. Section A and C carry **fifteen (15)** marks each and section B carries **seventy (70)** marks.
4. All writing must be in **black** or **blue** ink **except** diagrams which must be in pencil.
5. Communication devices and any unauthorized materials are **not** allowed in the assessment room.
6. Write your **Examination Number** at every page.

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**FORM TWO AND FORM FOUR PRE NECTA SERIES
ISESE TIME TABLE FOR GROUP II
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ALL SUBJECTS	DATE
SERIES - 01	03 rd August 2026
SERIES - 02	10 th August 2026
SERIES - 03	17 th August 2026
SERIES - 04	24 th August 2026
SERIES - 05	31 st August 2026

NB# SERIES No.06 – No.10 (SEPTEMBER – OCTOBER)

SECTION A (15 Marks)

Answer **all** questions in this section.

1. Choose the most correct alternative from among the given options and write its letter in the space provided.

(i) Juma noticed a brown, flaky layer forming on his family's iron gate after three months of rainy weather. His elder sister told him this is a chemical change known as rusting. Which observation best proves that a chemical change occurred?

- A. The iron changed its physical position.
- B. A new substance with entirely different properties was formed permanently.
- C. The gate can easily be restored to its original state by washing it with water.
- D. The total mass of the gate completely vanished into the air.

(ii) A laboratory technician accidentally spilled a concentrated chemical solution on his hand. Before running for medical help, what is the most immediate first aid action he should perform?

- A. Apply thick grease or cooking oil to the skin.
- B. Neutralize it immediately with a strong base or acid.
- C. Wash the affected area with plenty of running water for several minutes.
- D. Cover the area with a tight plastic wrapper.

(iii) During a laboratory session, a student adjusted the air hole of a Bunsen burner to be completely closed. Which type of flame will be produced, and what is its main characteristic?

- A. Non-luminous flame, which is very hot and clean.
- B. Luminous flame, which is yellow, produces soot, and is unsteady.
- C. Non-luminous flame, which produces a loud roaring sound.
- D. Luminous flame, which is blue and has three distinct zones.

(iv) A sample of matter has a definite volume but takes the shape of the container it is poured into. The particles are close together but can slide past one another. In which state of matter is this sample?

- A. Solid state
- B. Gas state
- C. Liquid state
- D. Plasma state

(v) Mariam was given a mixture containing fine sand and common salt. She wants to obtain pure crystals of salt. Which sequence of separation methods is the most appropriate?

- A. Evaporation, then sublimation, then filtration.
- B. Dissolving in water, filtration, then evaporation of the filtrate.
- C. Decantation, followed by magnetic separation.

D. Simple distillation, followed by fractional distillation.

(vi) An atom of an element has an atomic number of 11 and a mass number of 23. What is the correct composition of the subatomic particles located inside its nucleus?

- A. 11 protons and 11 electrons
- B. 11 protons and 12 neutrons
- C. 12 protons and 11 neutrons
- D. 23 protons and 11 electrons

(vii) An element X has an electronic configuration of 2:8:7. In which Group and Period of the Periodic Table is this element located?

- A. Group II, Period 7
- B. Group VII, Period 2
- C. Group VII, Period 3
- D. Group III, Period 7

(viii) Two atoms react together. Atom A (a metal) completely transfers its valence electron to atom B (a non-metal), resulting in the formation of a crystalline compound. What type of chemical bond is formed between them?

- A. Covalent bond
- B. Ionic bond
- C. Metallic bond
- D. Dative bond

(ix) A student dipped a blue litmus paper into an unknown clear solution Y. The blue litmus paper immediately turned red. Which of the following represents the most likely pH value for solution Y?

- A. 12.0
- B. 7.0
- C. 2.5
- D. 9.5

(x) When a small piece of sodium metal is dropped into a beaker containing cold water, a vigorous reaction occurs producing a gas that burns with a "pop" sound. What gas is evolved during this reaction?

- A. Oxygen
- B. Carbon dioxide
- C. Hydrogen
- D. Nitrogen oxide

ANSWER

i	ii	iii	iv	v	vi	vii	viii	ix	x

2. Match the chemical descriptions or processes in **List A** with their corresponding terms in **List B** by writing the letter of the correct response.

List A	List B
(i) A pure substance made up of two or more different elements chemically combined together in a fixed ratio.	A. Atom
(ii) The chemical reaction where an acid and a base react quantitatively to form a salt and water only.	B. Covalent Bonding
(iii) The standard temperature at which a liquid completely changes into its gas phase at atmospheric pressure.	C. Boiling Point
(iv) A chemical bond formed by the mutual sharing of valence electrons between non-metal atoms.	D. Compound
(v) The horizontal row of elements in the periodic table that indicates the number of occupied electron shells.	E. Mixture
	F. Neutralization
	G. Period
	H. Group

ANSWER

i	ii	iii	iv	v

SECTION B (70 Marks)

Answer **all** questions in this section.

3. In an industrial town, a new cosmetic manufacturing plant was opened. The local community is divided; some believe Chemistry is dangerous and creates toxic waste, while others argue that it drives modern development through the production of medicine, soap, and fertilizers.

(i) Explain why Chemistry is defined as the study of matter.

(ii) State two professional career fields where a deep knowledge of Chemistry is directly applied.

(iii) How does the chemical application of fertilizers benefit the agricultural sector in Tanzania?

(iv) State one negative effect of bad chemical waste disposal on the environment.

(v) Differentiate between a chemical product and a chemical raw material.

4. During a practical session, Form Two students were told to heat a substance that gives off toxic fumes. One student rushed to use a simple open wooden bench, while the teacher stopped him and pointed out specific specialized apparatus and regulatory safety stations designed for such experiments.

(i) Name the specialized safety chamber inside a chemistry laboratory where experiments involving toxic gases must be conducted.

(ii) State the function of a safety goggle and a pair of laboratory gloves during a practical session.

(iii) Why is it strictly forbidden to smell or taste unknown chemicals directly in the laboratory?

(iv) Identify two standard pieces of apparatus used for measuring fixed volumes of liquids accurately

(v) What does a warning sign showing a skeleton head and crossbones signify on a chemical container?

5. A welder uses a gas torch for cutting steel sheets. He notices that when he adjusts the torch valves to provide an excess supply of oxygen, the flame becomes pale blue, very hot, and quiet. However, when the oxygen supply is limited, the flame changes behavior entirely, producing a bright color and soot that covers the metal.

(i) Identify the type of flame produced when there is an excess supply of oxygen.

(ii) State the name of the hottest zone found in the pale blue flame.

(iii) Why is the soot-producing flame undesirable for heating glass apparatus in a laboratory?

(iv) Name the two chemical elements that must be present in the atmosphere for a standard fuel to burn and produce a flame.

(v) State the main difference between a luminous flame and a non-luminous flame in terms of structural zones.

6. A mining company extracts raw materials from the ground. They process a material that consists of different substances physically joined together in varying proportions, which can be separated easily. In another section, they deal with pure substances that cannot be broken down into simpler forms by any chemical means.

(i) Identify which material represents a mixture and which represents an element.

(ii) State two differences between a compound and a mixture.

(iii) What physical property is utilized when separating components using fractional distillation?

(iv) Why is the separation of iodine from sand achieved through sublimation?

(v) Give one domestic example of a homogeneous mixture found in a typical household.

7. A physics student argues that atoms are completely solid spheres without internal structures. A chemistry student corrects him by explaining that an atom is composed of a dense central core surrounded by subatomic particles moving in specific paths, which determines how elements interact and are arranged in the periodic table.

(i) Name the three subatomic particles that make up an atom.

(ii) State the location and the electrical charge of an electron.

(iii) What is meant by the term "Isotopes"?

(iv) If an element has 8 protons and 10 neutrons, write down its mass number.

(v) Why are elements placed in the same Group of the Periodic Table expected to exhibit similar chemical properties?

8. In a laboratory demonstration, a teacher shows two bottles. One contains Sodium Chloride (NaCl), which has a high melting point and conducts electricity when dissolved in water. The second contains Carbon Tetrachloride (CCl₄), which is a liquid at room temperature and does not conduct electricity.

(i) What type of chemical bonding exists in Sodium Chloride (NaCl)?

(ii) What type of chemical bonding exists in Carbon Tetrachloride (CCl₄)?

(iii) Explain why ionic compounds like sodium chloride have high melting points.

(iv) Why can ionic compounds conduct electricity in an aqueous state but not in a solid state?

(v) State the electronic condition required for an atom to achieve stability (duplet or octet rule).

9. A farmer tests his soil and finds it is too acidic to grow maize successfully. An agricultural officer advises him to spread slaked lime (Calcium hydroxide) over the farm. After a few weeks, the soil pH returns to normal, and a white crust forms on the surface due to salt formation.

(i) What name is given to the type of chemical reaction that occurred between the acidic soil and the slaked lime?

(ii) Define the term "Acid" according to the Arrhenius concept.

(iii) State two general properties of basic substances.

(iv) Differentiate between a normal salt and an acidic salt.

(v) Name one commercial use of salts in industries other than seasoning food.

SECTION C (15 Marks)

Answer **question number ten** from this section.

10. Chemical reactions are constantly occurring around us, inside our bodies, and in manufacturing plants. These reactions can be classified into different types depending on how the reactants interact to form products. Discuss the concept of chemical reactions by addressing the following areas:

- The definition of a chemical reaction and three indicators that show a chemical reaction has taken place.
- The difference between a combination reaction and a decomposition reaction, providing one real-life or industrial example for each.
- Two major factors that can change or accelerate the rate of a chemical reaction.

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